

MOMENT CLOSED FORM AND RH CRITERION

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$$\theta(t) = \arg \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{\log \pi}{2} t$$

$$\Theta(t) = \theta(t) + Ct, \quad C \in \mathbb{R} \text{ such that } \Theta'(t) > 0 \text{ for all } t \geq 0$$

$$\Theta_k := \Theta^{(k)}(0), \quad \Theta_1 = \frac{1}{2} \psi\left(\frac{1}{4}\right) - \frac{\log \pi}{2} + C$$

(all $\Theta_{2\ell} = 0$.)

$$\zeta_a := \zeta^{(a)}\left(\frac{1}{2}\right), \quad \varepsilon \in \mathbb{R}, \quad \varepsilon > 0$$

$$h(t) = \zeta\left(\frac{1}{2} + it\right) \sqrt{\Theta'(t)} (1 + \Theta(t)^2)^{-\varepsilon} e^{i\Theta(t)}$$

$$d\Phi_G(u) = \frac{h(t(u))}{\Theta'(t(u))} du, \quad t(u) = \Theta^{-1}(u)$$

$$Z(t) = e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right)$$

$$Z(t) = e^{-iCt} (\Theta'(t))^{1/2} (1 + \Theta(t)^2)^\varepsilon \int_{\mathbb{R}} \delta(u - \Theta(t)) d\Phi_G(u)$$

$$\mu_n = \int_{\mathbb{R}} u^n d\Phi_G(u) = \int_{-\infty}^{\infty} \Theta(t)^n \zeta\left(\frac{1}{2} + it\right) \sqrt{\Theta'(t)} (1 + \Theta(t)^2)^{-\varepsilon} e^{i\Theta(t)} dt$$

$$\mu_n = \frac{(-i)^n n!}{\Theta_1^{n+\frac{1}{2}}} \sum_{\substack{a,b,c \geq 0 \\ (k_3, k_5, \dots) \geq 0 \\ a+b+2c+\sum_{\ell \geq 1} 2\ell k_{2\ell+1} = n}} \frac{i^a \zeta_a}{a!} \cdot \frac{i^b}{b!} M_b \cdot \frac{\prod_{j=0}^{c-1} (-\varepsilon - j)}{c!} N_{2c} \cdot \frac{\prod_{j=0}^{K-1} (-(n+1) - j)}{K!} \prod_{\ell \geq 1} \frac{1}{k_{2\ell+1}!} \left(\frac{\Theta_{2\ell+1}}{(2\ell+1)! \Theta_1} \right)^{k_{2\ell+1}}$$

$$K = \sum_{\ell \geq 1} k_{2\ell+1}$$

$$M_b = \sum_{\substack{(m_1, m_3, \dots) \geq 0 \\ \sum_{\ell \geq 0} (2\ell+1) m_{2\ell+1} = b}} \frac{b!}{\prod_{\ell \geq 0} m_{2\ell+1}!} \prod_{\ell \geq 0} \left(\frac{\Theta_{2\ell+1}}{(2\ell+1)!} \right)^{m_{2\ell+1}}$$

$$N_{2c} = \sum_{\substack{(p_1, p_3, \dots) \geq 0 \\ \sum_{\ell \geq 0} (2\ell+1) p_{2\ell+1} = 2c}} \frac{(2c)!}{\prod_{\ell \geq 0} p_{2\ell+1}!} \prod_{\ell \geq 0} \left(\frac{\Theta_{2\ell+1}}{(2\ell+1)!} \right)^{p_{2\ell+1}}$$

$$\Psi(z) = \sum_{n=0}^{\infty} \frac{\mu_n \sqrt{\Theta_1}}{n!} z^n, \quad A = C - \frac{\log \pi}{2}$$

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$$Z_\Psi(r) = \{z \in \mathbb{C} : |z| < r, \Psi(z) = 0\}, \quad \Gamma(r) = \{\gamma \in \mathbb{R} : |\gamma| < r, \zeta(\tfrac{1}{2} + i\gamma) = 0\}$$

The Riemann Hypothesis is the statement

$$Z_\Psi(r) = \Gamma(r) \quad \text{for every } r \in (0, 1/A),$$

equivalently

$$\frac{1}{2\pi} \int_0^{2\pi} \log |\Psi(re^{i\varphi})| d\varphi - \log |\Psi(0)| = \sum_{\gamma \in \Gamma(r)} \log \frac{r}{|\gamma|} \quad \text{for every } r \in (0, 1/A).$$

REMARK ON THE ROLE OF $\sqrt{\Theta'(t)}$

Since $h(t) = e^{iCt} \sqrt{\Theta'(t)} (1 + \Theta(t)^2)^{-\varepsilon} Z(t)$, one has

$$|h(t)| = \sqrt{\Theta'(t)} (1 + \Theta(t)^2)^{-\varepsilon} |Z(t)|,$$

so pointwise h is *larger* than Z by the factor $\sqrt{\Theta'(t)} \sim \sqrt{(\log t)/2}$, yet $h \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ for every $\varepsilon > 0$ while Z is not integrable.

The $\sqrt{\Theta'}$ prefactor is the Riemann–von Mangoldt Jacobian: it rescales t -arclength to Θ -arclength, under which the zeros of $\zeta(\frac{1}{2} + it)$ become asymptotically uniformly distributed in the variable $u = \Theta(t)$ with density $1/(2\pi)$. The pushforward measure Φ_G therefore lives on $\mathbb{R}$ with uniform zero density, and the damping $(1 + u^2)^{-\varepsilon}$ is exactly what the uniform spacing can tolerate to yield finite moments of every order.

Without the $\sqrt{\Theta'}$ compensation, damping Z directly by $(1 + t^2)^{-\varepsilon}$ suppresses amplitude faster than the zero density $(\log t)/(2\pi)$ accumulates; the damped function would run out of oscillation and lose the spectral information about ζ . The combined weight $\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon}$ is the unique balance that both (i) makes h integrable, and (ii) preserves the infinite oscillation needed to encode the zeros.